1.1 Interval Notation and Set Notation

Name _____ Period ____ Date _____

Algebra II 1.1 · Student sheet

I can…

- write inequalities and graphs in interval notation.
- write sets of numbers in set-builder notation.
- graph sets of numbers and use the notations for real-life situations.

Vocabulary · fill in the missing word

Interval notation — writes a set of numbers using its _____, such as [−2, 5)

[ ] — the _____ is included (≤ or ≥, closed circle)

( ) — the _____ is not included (< or >, open circle); always use ( ) with ∞ and −∞

Set-builder notation — {x | _____}, read "the set of all x such that …"

Union ∪ — combines two _____ ("or")

Part 1 · Match-up

Write the letter of the matching graph and the matching interval for each inequality.

Graphs: A. open at −1, shaded left B. closed at 0 and 6 C. whole line shaded D. closed at 3, shaded right E. open at −4, closed at 2 F. closed at −2 shaded left, open at 1 shaded right

Intervals: P. (−4, 2] Q. (−∞, ∞) R. [3, ∞) S. (−∞, −2] ∪ (1, ∞) T. [0, 6] U. (−∞, −1)

Inequality	Graph	Interval
$x \geq 3$		
$x < -1$		
$-4 < x \leq 2$		
$0 \leq x \leq 6$		
$x \leq -2$ or $x > 1$		
all real numbers		

Part 2 · Notes and examples (fill in each note, then work the example)

Brackets and parentheses: A bracket [ ] means the endpoint is _____; a parenthesis ( ) means it is _____.

Infinity: ∞ and −∞ always use a _____.

Example 1

Write each inequality in interval notation.

a. $x > 4$ b. $-3 \leq x < 7$ c. $x \leq 0$ or $x \geq 5$

Example 2

A number line has a closed circle at −2 and an open circle at 3, with the segment between them shaded. Write the set in interval notation.

Set-builder notation: Set-builder notation: {x | _____} is read “the set of all x _____.”

Example 3

Write each interval in set-builder notation.

a. (−5, 1] b. [0, ∞)

Example 4

A roller coaster requires riders to be at least 48 inches tall and less than 76 inches tall. Write the allowed heights in interval notation and in set-builder notation.

Part 3 · Partner practice

Partner 1

Write each in interval notation.

a. $x \le -6$ b. $-1 < x < 8$

Partner 2

Write $(-\infty, 3) \cup [9, \infty)$ in set-builder notation.